

# Ana Lorena Ortiz Loyola

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Data science student (Tecnológico de Monterrey, expected 2027) who uses Claude Code, an AI coding assistant, as a daily engineering tool, treating prompting as an engineering discipline rather than a one-off exercise. Combines this with hands-on automation experience at Deutsche Bundesbank, building Python-based automated testing, error detection, and automated ticketing/notification workflows, plus a machine learning foundation (Scikit-Learn, Keras, TensorFlow). Works in a structured, documentation-first way suited to building and scaling AI use cases.

## Core Competencies

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- **Applied AI & Prompt Engineering:** Daily hands-on use of Claude Code on a real end-to-end project; treats prompting, workflow design, and documentation of AI use cases as an engineering discipline
- **Automation & Workflow Engineering:** Python/Pandas-based automated testing and error detection, with automated Jira ticketing and notification workflows (Deutsche Bundesbank AnaCredit)
- **Machine Learning:** Scikit-Learn, Keras, TensorFlow, Random Forest, SVM, K-Means
- **Programming:** Python, R, SQL, Java, C++
- **Documentation & Enablement:** Structured technical documentation, playbooks, and clear stakeholder communication
- **Cloud & Tools:** Azure Cognitive Services, AWS, Git, REST APIs

## Professional Experience

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- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**  
*Data Engineer / Intern* *04/2026–07/2026*
  - Built automated Python/Pandas testing (row counts, referential integrity, aggregate consistency) to validate business logic after data migration
  - Designed automated error detection for large-scale reported banking data, with automated Jira ticket creation and stakeholder emails, cutting manual review effort and response time to data quality issues

## Projects

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- **Personal Project (GitHub), built with Claude Code**  
*Banking Data Pipeline – Data Cleaning & Audit Log* 05/2026
  - Designed and built an end-to-end data pipeline for simulated bank transactions (150 accounts), using Claude Code as an AI coding assistant throughout: prompt design, iterative testing, and documentation
  - Built a normalized SQLite database with foreign-key integrity checks, staged data cleaning, and a fully auditable log recording every change
  - Produced quality visualizations and a structured exception report
- **Tecnológico de Monterrey**  
*Predictive Modeling – Healthcare Optimization* 10/2024–12/2024
  - Built and validated predictive models (Random Forest, Neural Networks) and K-Means clustering on 25+ years of national health data

## Education

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- **Tecnológico de Monterrey** Mexico  
*BSc in Data Science* 2022–2027  
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: machine learning & AI, data analysis & statistics, numerical optimization.
- **Universität Göttingen** Germany  
*Exchange Semester* 2025–2026

## Languages

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- German (C1), English (C1, TOEFL iBT 105/120), Spanish (native).

## Publications

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- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

## Honors and Awards

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- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

## References

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- Available upon request.